research_paper_codeabstract

CodeAbstract: Privacy-Preserving AST-Based Semantic Abstraction for Secure LLM Code Consultation

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Abstract

Large Language Models (LLMs) have become indispensable for code review, debugging, and optimization, yet submitting proprietary source code to cloud-hosted LLM endpoints exposes organizations to severe intellectual property (IP) leakage risks—including exposure of API keys, credentials, proprietary algorithms, and business-critical constants. We present **CodeAbstract**, a client-side, AST-based semantic abstraction framework that transforms Python source code into a privacy-safe intermediate representation before cloud LLM consultation, then deterministically reconstructs the LLM's feedback into the developer's original code context. CodeAbstract operates through four formally specified algorithms: ABSTRACT, CLASSIFY, RECONSTRUCT, and ENCRYPT. We evaluate CodeAbstract across five metric pillars on a curated benchmark corpus of 10 diverse Python modules (390 LOC, 11 code domains) and a hand-labeled dataset of 150 AST nodes. Our empirical results demonstrate: (1) a Privacy Information Ratio (PIR) of **0.0024** with **0.0% secret leakage**; (2) **100.0% reconstruction token hit rate**; (3) classification Macro-F1 of **0.8937**; (4) **perfect AST isomorphism** (1.0000) with zero cyclomatic complexity change; and (5) throughput of **9,872 LOC/sec** on commodity CPU hardware. Comparative analysis against four state-of-the-art frameworks—ECCC (ACL W-NLP 2025), De Tomasi et al. (ACM EASE 2025), CodeCipher (2024), and MaskSQL (arXiv 2025)—demonstrates that CodeAbstract achieves superior or competitive privacy protection while uniquely guaranteeing deterministic, lossless reconstruction and complete structural invariance.

Keywords: Code privacy, AST abstraction, LLM security, sensitive token classification, deterministic reconstruction, privacy-preserving code review

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1. Introduction

1.1 Motivation

The rapid adoption of cloud-hosted Large Language Models (LLMs) such as GPT-4 (OpenAI, 2023), Gemini (Google DeepMind, 2024), and Claude (Anthropic, 2024) for software engineering tasks—including automated code review, bug detection, security auditing, and code translation—has created a fundamental tension between **utility** and **privacy**. Developers routinely paste proprietary code snippets into commercial LLM endpoints, inadvertently exposing:

- **Credentials and secrets**: API keys, database connection strings, OAuth tokens, JWT signing keys
- **Business logic**: Pricing algorithms, commission rates, discount formulas, revenue calculations
- **Infrastructure details**: Internal endpoints, server hostnames, routing topologies, database schemas
- **Customer data**: User identifiers, account references, PII-adjacent variables

According to Samsung's 2023 internal audit, three separate incidents of proprietary source code leakage through ChatGPT were detected within a single month, prompting a company-wide ban on generative AI tools (TechCrunch, 2023). GitGuardian's 2024 State of Secrets Sprawl report identified over 12.8 million hardcoded secrets exposed in public GitHub repositories, underscoring the systemic nature of credential leakage in modern development workflows.

1.2 Problem Statement

The core challenge is formalized as follows: Given source code S containing sensitive tokens $T_s \subset T$ (where T is the set of all tokens), and a cloud-hosted LLM function $\mathcal{L}: S \rightarrow R$ that produces a review response R, design a transformation pipeline that:

1. **Privacy Guarantee**: Produces an abstracted representation $\tilde{S}$ such that $T_s \cap \text{tokens}(\tilde{S}) = \emptyset$
2. **Utility Preservation**: The LLM can still produce useful feedback: $\mathcal{L}(\tilde{S}) \approx \mathcal{L}(S)$ in semantic quality
3. **Lossless Reconstruction**: A local reconstruction function $\mathcal{R}$ deterministically recovers: $\mathcal{R}(\mathcal{L}(\tilde{S}), \tilde{M}, K) \rightarrow F$ where F maps all abstract tokens back to original identifiers
4. **Structural Invariance**: $\text{AST}(\tilde{S}) \cong \text{AST}(S)$ (isomorphic abstract syntax trees)

1.3 Contributions

This paper makes the following contributions:

1. **A formal four-algorithm pipeline** (ABSTRACT, CLASSIFY, RECONSTRUCT, ENCRYPT) for AST-level semantic abstraction with provably deterministic, bijective reconstruction
2. **A multi-class sensitivity classifier** achieving Macro-F1 of 0.8937 across SECRET, VALUE, and IDENTIFIER categories on a curated 150-node ground-truth dataset spanning 11 code domains
3. **A five-pillar evaluation framework** measuring Privacy Information Ratio (PIR), Reconstruction Fidelity, Classification Accuracy, Structural Preservation, and Performance, reproducible via automated harness
4. **Comprehensive comparative analysis** against four state-of-the-art privacy-preserving code frameworks, demonstrating superior privacy-utility tradeoffs

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2. Related Work

2.1 Edge-Cloud Code Completion (ECCC)

He et al. (2025) propose ECCC, an edge-cloud collaborative framework for privacy-preserving code completion. ECCC deploys a small language model (SLM) locally to handle sensitive completions while routing non-sensitive queries to cloud LLMs. Their approach uses LibCST-based AST token substitution with cloaked docstrings and an ephemeral RAM mapping table $\mathcal{M}$.

Reported metrics: Privacy Score = 10.0/10.0, Pass@1 = 90.0% (HumanEval), Functional Score = 8.93/10.0.

Limitations: Requires a dedicated GPU with ≥16 GB VRAM for the local 4-bit MoE inference model; the privacy evaluation is manual (human-scored on a 0–10 scale) rather than automated; and the approach does not provide formal structural invariance guarantees.

2.2 Simplicity by Obfuscation (De Tomasi et al.)

De Tomasi et al. (2025) evaluate LLM-driven code obfuscation transformations at ACM EASE 2025. They prompt GPT-4, Claude 3.5, and Gemini 1.5 to obfuscate code and measure the resulting **Semantic Elasticity (SE)**:

$$
SE = \frac{|\Delta CC| \times P^2}{E}
$$

where ΔCC is the cyclomatic complexity change, P is the functional pass rate, and $E = \frac{\text{LOC}_{\text{obfuscated}}}{\text{LOC}_{\text{original}}}$ is the code expansion ratio.

Reported metrics: Best Pass Rate $P = 80.93\%$ (GPT-4 few-shot), $\Delta CC = -1.27$ (Claude 3.5), SE range $[0.182, 0.730]$.

Limitations: Obfuscation is one-way (no reconstruction); pass rate drops to 28.55% under zero-shot; complexity changes indicate semantic mutation rather than preservation; and the approach is non-deterministic (LLM-generated transformations vary between runs).

2.3 CodeCipher

Lin et al. (2024) propose CodeCipher, which perturbs continuous token embeddings to prevent identifier recovery while maintaining task performance. The approach operates in the embedding space, applying learned permutations to token representations before cloud inference.

Reported metrics: Pass@1 = 47.56% (vs. 50.60% unperturbed), Deobfuscation Rate = 34.0%, Compilation Rate = 0% (perturbed code is not valid syntax).

Limitations: Operates at the embedding level, requiring white-box access to model internals; produces syntactically invalid code (0% compilation rate); and the 34% deobfuscation rate indicates significant residual privacy leakage.

2.4 MaskSQL

Abedini et al. (2025) present MaskSQL, a bijective token-masking framework for privacy-preserving Text-to-SQL. MaskSQL replaces schema names and literal values with masked placeholders before sending queries to an untrusted cloud LLM, then uses a local SLM (Qwen2.5-7B) for post-reconstruction correction.

Reported metrics: Execution Accuracy (EX) = 55.66% (Ψ_F) to 62.66% (Ψ_C), Masking Recall (MR) = 61.36%, Re-identification Score (RI) = 75.47%.

Limitations: Domain-specific to SQL (not general-purpose code); requires a local 7B-parameter SLM for reconstruction correction; masking recall of 61.36% means 38.64% of sensitive tokens remain exposed; and latency of 150–300 ms per query limits real-time IDE integration.

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3. System Architecture and Methodology

3.1 Overview

CodeAbstract implements a four-stage pipeline that operates entirely on the client side, with only the sanitized intermediate representation $\tilde{S}$ transmitted to the cloud LLM:

$$
S \xrightarrow{\text{ABSTRACT}} (\tilde{S}, \tilde{M}, K) \xrightarrow{\text{Cloud LLM}} R \xrightarrow{\text{RECONSTRUCT}} F
$$

where:

- S = original source code
- $\tilde{S}$ = abstracted (privacy-safe) source code
- $\tilde{M}$ = encrypted mapping table (AES-128-CBC + HMAC-SHA256)
- K = local encryption key (never transmitted)
- R = LLM response containing abstract tokens
- F = final reconstructed feedback with original identifiers

3.2 Algorithm 1: ABSTRACT(S, k)

The abstraction algorithm parses Python source code into an AST, traverses all nodes, classifies each for sensitivity, generates unique typed tokens with consistent reuse, and unparses back to abstract source code.

Input: Source code S, context window size k, privacy level $\ell \in \{\text{low}, \text{medium}, \text{high}, \text{critical}\}$

Output: Abstracted code $\tilde{S}$, encrypted mapping $\tilde{M}$, encryption key K

```
Algorithm 1: ABSTRACT(S, k, ℓ)
──────────────────────────────────────────────────
1.  T ← AST.parse(S)                              // Parse to AST
2.  M ← ∅                                         // Initialize mapping table
3.  C ← {FUNC: 0, VAR: 0, SECRET: 0, ...}         // Token counters
4.  seen ← ∅                                       // Name→token cache
5.  for each node e ∈ AST.walk(T) do
6.      if ¬is_significant(e) then continue
7.      name ← extract_name(e)
8.      label ← CLASSIFY(e)                        // Algorithm 2
9.      if ¬should_abstract(label, ℓ) then continue
10.     if name ∈ seen then
11.         token ← seen[name]                     // Reuse for consistency
12.     else
13.         prefix ← label_to_prefix(label, type(e))
14.         token ← prefix + "_" + C[prefix]++
15.         ctx ← GET_CONTEXT(S, e.lineno, k)
16.         M[token] ← {name, label, type(e), lineno, ctx}
17.         seen[name] ← token
18.     replace_node(e, token)
19. S̃ ← AST.unparse(T)                             // Unparse back to code
20. (M̃, K) ← ENCRYPT(M)                           // Algorithm 4
21. return (S̃, M̃, K)
```

Key Properties:

- **Bijective mapping**: Each unique identifier maps to exactly one token, and each token maps back to exactly one identifier ($|M|$ is a bijection)
- **Consistent reuse**: If the same name appears multiple times in the AST, the same token is reused (Line 10–11), ensuring referential consistency in $\tilde{S}$
- **AST-level operation**: Replacement occurs at the AST node level, not via string substitution, guaranteeing syntactic validity of $\tilde{S}$

3.3 Algorithm 2: CLASSIFY(e)

The classification algorithm assigns each AST node a sensitivity label from the set $\mathcal{L} = \{\text{SECRET}, \text{VALUE}, \text{IDENTIFIER}\}$.

```
Algorithm 2: CLASSIFY(e)
──────────────────────────────────────────────────
1.  text ← extract_text(e)                        // Name + literal value
2.  if matches_any(text, SECRET_PATTERNS) then
3.      return SECRET
4.  if matches_any(text, VALUE_PATTERNS) then
5.      return VALUE
6.  if is_numeric_constant(e) then
7.      return VALUE
8.  return IDENTIFIER
```

Classification hierarchy (decreasing sensitivity):

- **SECRET**: Credentials, API keys, tokens, passwords, connection strings (matched by 14 regex patterns including `api[_-]?key`, `password`, `secret`, `token`, `credential`, etc.)
- **VALUE**: Business-sensitive literals—prices, rates, endpoints, URLs, email addresses, numeric constants
- **IDENTIFIER**: All other semantically significant names (functions, variables, classes, parameters)

3.4 Algorithm 3: RECONSTRUCT(R, $\tilde{M}$, K)

The reconstruction algorithm deterministically replaces all abstract tokens in the LLM response with original identifiers and appends formatted source context.

```
Algorithm 3: RECONSTRUCT(R, M̃, K)
──────────────────────────────────────────────────
M ← DECRYPT(M̃, K)                          // Algorithm 4
tokens ← extract_tokens(R)                  // Regex:
(FUNC|VAR|SECRET|CONST|NUM|CLASS)_\d+
F ← R                                       // Working copy
refs ← ∅                                    // Reference list
for each token t ∈ tokens do
    if t ∉ M then continue
    entry ← M[t]
    F ← F.replace(t, entry.name)            // Substitute back
    refs ← refs ∪ {entry}                   // Collect references
F ← F + format_context_blocks(refs)         // Append context
return F
```

Correctness guarantee: Since ABSTRACT creates a bijective mapping $M: \text{token} \leftrightarrow \text{name}$ and RECONSTRUCT applies the inverse mapping M^{-1}, every token in R that exists in M is deterministically resolved:

$$
\forall t \in \text{tokens}(R) \cap \text{dom}(M): \quad \mathcal{R}(t) = M^{-1}(t) = \text{original_name}
$$

3.5 Algorithm 4: ENCRYPT/DECRYPT

The mapping table is secured using Fernet symmetric encryption (AES-128-CBC with HMAC-SHA256 authentication):

$$
\text{ENCRYPT}(M) = (\text{Fernet}_K(\text{JSON}(M)),\ K)
$$

$$
\text{DECRYPT}(\tilde{M}, K) = \text{JSON}^{-1}(\text{Fernet}_K^{-1}(\tilde{M}))
$$

where K is a randomly generated 256-bit Fernet key that **never leaves the local machine**.

3.6 Privacy Level Hierarchy

CodeAbstract implements a four-tier privacy hierarchy that controls abstraction aggressiveness:

Level	Abstracted Categories	Use Case
`low`	SECRET only	Minimal masking; credentials only
`medium`	SECRET + VALUE	Standard; protects credentials and business values
`high`	SECRET + VALUE + IDENTIFIER	Full privacy; all identifiers abstracted
`critical`	All nodes	Maximum; complete code anonymization

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4. Evaluation Methodology

4.1 Benchmark Corpus

We evaluate CodeAbstract on a curated corpus of **10 Python source modules** spanning **11 code domains** with a total of **390 lines of code** (LOC). The corpus is designed to cover diverse sensitivity profiles:

Module	LOC	Domain	Secrets	Values	Identifiers
`api_client`	42	API Integration	10	16	—
`ecommerce_pricing`	39	E-Commerce	0	82	—
`ml_pipeline`	48	Machine Learning	5	21	—
`web_server_config`	31	Web Infrastructure	11	16	—
`simple_utility`	41	Utility Functions	0	9	—
`auth_service`	42	Authentication	25	9	—
`financial_calculator`	33	Finance	0	40	—
`database_layer`	41	Database	6	9	—
`microservice_router`	36	Microservices	3	28	—
`analytics_engine`	37	Analytics	5	37	—
Total	**390**	**11 domains**	**65**	**267**	—

4.2 Ground-Truth Labeled Dataset

For classification evaluation (Pillar 3), we construct a hand-labeled dataset of **150 AST nodes** (50 per class) spanning 11 code domains. Each node is annotated with:

- `node_name`: The identifier string
- `node_type`: AST node type (Name, FunctionDef, ClassDef, Constant, arg, Attribute)
- `expected_label`: Ground-truth classification (SECRET, VALUE, IDENTIFIER)
- `domain`: Code domain for stratified analysis

4.3 Experimental Setup

All experiments are conducted on a consumer-grade machine:

- **CPU**: Intel Core i5 (no GPU required)
- **RAM**: 8 GB
- **OS**: Windows 10/11
- **Python**: 3.12
- **Privacy Level**: `high` (all categories abstracted)
- **Repetitions**: 5 per timing measurement (median reported)

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5. Evaluation Results and Mathematical Proofs

5.1 Pillar 1: Privacy Information Ratio (PIR)

5.1.1 Definition

The Privacy Information Ratio measures the fraction of sensitive tokens that survive abstraction (i.e., leak into the abstracted output):

$$
\text{PIR} = 1 - \frac{N_{\text{abstracted}}}{N_{\text{sensitive}}}
$$

where:

- $N_{\text{sensitive}} = |T_{\text{SECRET}}| + |T_{\text{VALUE}}|$ is the total count of sensitive tokens (secrets + values) in the original source
- $N_{\text{abstracted}}$ is the count of those tokens successfully replaced with abstract placeholders

Interpretation: $\text{PIR} = 0.0$ means perfect privacy (all sensitive tokens abstracted); $\text{PIR} = 1.0$ means no privacy (no tokens abstracted).

The per-category leakage rates are defined as:

$$
\text{SecretLeakage} = \frac{|\{t \in T_{\text{SECRET}} : t \in
\text{tokens}(\tilde{S})\}|}{|T_{\text{SECRET}}|}
$$

$$
\text{ValueLeakage} = \frac{|\{t \in T_{\text{VALUE}} : t \in
\text{tokens}(\tilde{S})\}|}{|T_{\text{VALUE}}|}
$$

Leakage detection uses word-boundary regex matching: a token is considered “leaked” if it appears verbatim (as a whole word) in the abstracted output $\tilde{S}$.

5.1.2 Empirical Results

Module	Secret Total	Secret Abstracted	Value Total	Value Abstracted	PIR
`api_client`	10	10	16	16	**0.0000**
`ecommerce_pricing`	0	0	82	80	**0.0244**
`ml_pipeline`	5	5	21	21	**0.0000**
`web_server_config`	11	11	16	16	**0.0000**
`simple_utility`	0	0	9	9	**0.0000**
`auth_service`	25	25	9	9	**0.0000**
`financial_calculator`	0	0	40	40	**0.0000**
`database_layer`	6	6	9	9	**0.0000**
`microservice_router`	3	3	28	28	**0.0000**
`analytics_engine`	5	5	37	37	**0.0000**
Mean	**65**	**65**	**267**	**265**	**0.0024**

5.1.3 Proof of PIR

Computation:

$$
\text{Mean PIR} = \frac{1}{10} \sum_{i=1}^{10} \text{PIR}_i = \frac{1}{10}(0 + 0.0244 + 0 + 0 + 0 + 0 + 0 + 0 + 0 + 0) = \frac{0.0244}{10} = \mathbf{0.0024}
$$

Secret Leakage Rate:

$$
\text{SecretLeakage} = \frac{65 - 65}{65} = \frac{0}{65} = \mathbf{0.0\%}
$$

Value Leakage Rate:

$$
\text{ValueLeakage} = \frac{267 - 265}{267} = \frac{2}{267} = \mathbf{0.0075} \approx \mathbf{0.24\%}
$$

[!IMPORTANT]
The two leaked values in `ecommerce_pricing` are common numeric constants (e.g., `0`, `1`) that appear both as abstracted tokens and as Python built-in syntax elements. Secret leakage is **exactly zero** across all 10 modules.

Privacy Retention Score:

$$
\text{PRS} = 1 - \text{PIR} = 1 - 0.0024 = \mathbf{99.76\%}
$$

5.2 Pillar 2: Reconstruction Fidelity

5.2.1 Definition

Reconstruction fidelity measures how accurately the RECONSTRUCT algorithm resolves abstract tokens back to original identifiers. We define two sub-metrics:

Token Hit Rate — the fraction of abstract tokens in a simulated LLM response that are successfully looked up and replaced:

$$
\text{HitRate} = \frac{|\{t \in \text{tokens}(R) : t \in \text{dom}(M) \land M^{-1}(t) \in \text{tokens}(F)\}|}{|\text{tokens}(R) \cap \text{dom}(M)|}
$$

Roundtrip Fidelity — the character-level similarity between the original source S and the reconstructed output of a response containing all tokens:

$$
\text{RF} = \text{SequenceMatcher}(S, \mathcal{R}(\text{concat}(\text{dom}(M)), \tilde{M}, K))
$$

5.2.2 Empirical Results

Module	Tokens Generated	Token Hit Rate	Roundtrip Fidelity
`api_client`	47	**100.0%**	0.0654
`ecommerce_pricing`	47	**100.0%**	0.3113
`ml_pipeline`	64	**100.0%**	0.2000
`web_server_config`	41	**100.0%**	0.1892
`simple_utility`	44	**100.0%**	0.3757
`auth_service`	57	**100.0%**	0.2180
`financial_calculator`	46	**100.0%**	0.4354
`database_layer`	48	**100.0%**	0.3835
`microservice_router`	47	**100.0%**	0.1582
`analytics_engine`	52	**100.0%**	0.2857
Mean	**493 total**	**100.0%**	**0.2622**

5.2.3 Proof of Token Hit Rate

Theorem: For all tokens t generated by ABSTRACT and present in a response R, RECONSTRUCT resolves t to the correct original name with probability 1.

Proof: By construction of Algorithm 1 (ABSTRACT):

1. Every abstracted node produces a unique token $t_i = \text{prefix}_j$, stored in mapping $M[t_i] = \{\text{name}_i, \ldots\}$
2. The token format `(FUNC|VAR|SECRET|CONST|NUM|CLASS)_\d+` is deterministic and non-colliding (monotonic counter per prefix)
3. By Algorithm 3 (RECONSTRUCT), for each $t \in \text{tokens}(R) \cap \text{dom}(M)$: $F = F.\text{replace}(t, M[t].\text{name})$

Since M is a bijection (no two distinct names map to the same token, and no two tokens map to the same name), and string replacement is exact:

$$
\text{HitRate} = \frac{493}{493} = \mathbf{1.0000} = \mathbf{100.0\%} \quad \blacksquare
$$

Note on Roundtrip Fidelity: The roundtrip fidelity of 0.2622 reflects that reconstructing a *concatenation of all tokens* (not the original code structure) produces a different string arrangement than the original source. This is expected and does not indicate reconstruction error—it measures string-level similarity of an artificial roundtrip, not functional correctness.

5.3 Pillar 3: Classification Accuracy

5.3.1 Definitions

For each class $c \in \{\text{SECRET}, \text{VALUE}, \text{IDENTIFIER}\}$:

$$
\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}
$$

$$
\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}
$$

$$
F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}
$$

Macro-averaged metrics (unweighted mean across classes):

$$
\text{Macro-F1} = \frac{1}{|\mathcal{L}|} \sum_{c \in \mathcal{L}} F1_c
$$

5.3.2 Confusion Matrix and Per-Class Results

Confusion Matrix (rows = true label, columns = predicted label):

	Pred: SECRET	**Pred: VALUE**	**Pred: IDENTIFIER**
True: SECRET	**48**	1	1
True: VALUE	0	**44**	6
True: IDENTIFIER	1	7	**42**

Per-Class Metrics:

Class	**TP**	**FP**	**FN**	**Precision**	**Recall**	**F1**	**Support**
SECRET	48	1	2	**0.9796**	**0.9600**	**0.9697**	50
VALUE	44	8	6	**0.8462**	**0.8800**	**0.8628**	50
IDENTIFIER	42	7	8	**0.8571**	**0.8400**	**0.8485**	50
Macro	—	—	—	**0.8943**	**0.8933**	**0.8937**	**150**

5.3.3 Proof of Metrics

SECRET Precision

:

$$
P_{\text{SECRET}} = \frac{48}{48 + 1} = \frac{48}{49} = \mathbf{0.9796}
$$

SECRET Recall

:

$$
R_{\text{SECRET}} = \frac{48}{48 + 2} = \frac{48}{50} = \mathbf{0.9600}
$$

SECRET F1

:

$$
F1_{\text{SECRET}} = \frac{2 \times 0.9796 \times 0.9600}{0.9796 + 0.9600} = \frac{1.8808}{1.9396} = \mathbf{0.9697}
$$

VALUE Precision

:

$$
P_{\text{VALUE}} = \frac{44}{44 + 8} = \frac{44}{52} = \mathbf{0.8462}
$$

VALUE Recall

:

$$
R_{\text{VALUE}} = \frac{44}{44 + 6} = \frac{44}{50} = \mathbf{0.8800}
$$

VALUE F1

:

$$
F1_{\text{VALUE}} = \frac{2 \times 0.8462 \times 0.8800}{0.8462 + 0.8800} = \frac{1.4893}{1.7262} = \mathbf{0.8628}
$$

IDENTIFIER Precision

:

$$
P_{\text{ID}} = \frac{42}{42 + 7} = \frac{42}{49} = \mathbf{0.8571}
$$

IDENTIFIER Recall

:

$$
R_{\text{ID}} = \frac{42}{42 + 8} = \frac{42}{50} = \mathbf{0.8400}
$$

IDENTIFIER F1

:

$$
F1_{\text{ID}} = \frac{2 \times 0.8571 \times 0.8400}{0.8571 + 0.8400} = \frac{1.4399}{1.6971} = \mathbf{0.8485}
$$

Macro-F1

:

$$
\text{Macro-F1} = \frac{0.9697 + 0.8628 + 0.8485}{3} = \frac{2.6810}{3} = \mathbf{0.8937}
$$

Overall Accuracy

:

$$
\text{Accuracy} = \frac{48 + 44 + 42}{150} = \frac{134}{150} = \mathbf{0.8933} = \mathbf{89.33\%} \quad \blacksquare
$$

5.3.4 Domain-Stratified Accuracy

Domain	Accuracy	Domain	Accuracy
`api_client`	**100.0%**	`auth`	88.24%
`ml`	94.44%	`ecommerce`	82.86%
`microservice`	**100.0%**	`web`	80.00%
`crypto`	**100.0%**	`database`	83.33%
`analytics`	**100.0%**	`finance`	78.57%
`utility`	**100.0%**		

5.4 Pillar 4: Code Structure Preservation

5.4.1 Definitions

AST Isomorphism Score — measures structural preservation by comparing the distribution of structural AST node types between original and abstracted code:

$$
\text{ISO} = \frac{\sum_{n \in \mathcal{N}_s} \min(\text{freq}_{\text{orig}}(n), \text{freq}_{\text{abst}}(n))}{\sum_{n \in \mathcal{N}_s} \text{freq}_{\text{orig}}(n)}
$$

where $\mathcal{N}_s$ is the set of structural node types: {FunctionDef, AsyncFunctionDef, ClassDef, If, For, While, Try, ExceptHandler, Return, Assign, AugAssign, AnnAssign, Call, BoolOp, Compare, BinOp, With, Assert, Raise, Import, ImportFrom}.

Cyclomatic Complexity Delta — the absolute difference in McCabe's cyclomatic complexity:

$$
\Delta CC = |CC(\tilde{S}) - CC(S)|
$$

where $CC(S) = 1 + |\text{If}| + |\text{For}| + |\text{While}| + |\text{Except}| + \sum_{\text{BoolOp}} (|\text{operands}| - 1)$

Syntax Validity Rate — the fraction of abstracted files that parse as valid Python:

$$
\text{SVR} = \frac{|\{f : \text{ast.parse}(\tilde{S}_f) \text{ succeeds}\}|}{|F|}
$$

5.4.2 Empirical Results

Module	Syntax Valid	AST Isomorphism	ΔCC	Token Ratio	Abstraction Density
`api_client`	**Yes**	**1.0000**	**0**	0.9488	104.9%
`ecommerce_pricing`	**Yes**	**1.0000**	**0**	1.0270	127.0%
`ml_pipeline`	**Yes**	**1.0000**	**0**	0.9857	115.7%
`web_server_config`	**Yes**	**1.0000**	**0**	0.9818	101.8%
`simple_utility`	**Yes**	**1.0000**	**0**	0.8878	88.8%
`auth_service`	**Yes**	**1.0000**	**0**	0.9682	106.8%
`financial_calculator`	**Yes**	**1.0000**	**0**	1.0036	130.4%
`database_layer`	**Yes**	**1.0000**	**0**	0.9747	97.5%
`microservice_router`	**Yes**	**1.0000**	**0**	1.0864	118.6%
`analytics_engine`	**Yes**	**1.0000**	**0**	1.1486	114.9%
Mean	**100.0%**	**1.0000**	**0.00**	**0.9967**	**110.6%**

5.4.3 Proof of Structural Invariance

Theorem: The ABSTRACT algorithm preserves AST structural isomorphism: $\text{ISO}(S, \tilde{S}) = 1.0$.

Proof: ABSTRACT operates exclusively on leaf-level identifier nodes (Name, FunctionDef.name, ClassDef.name, arg.arg, Attribute.attr, Constant.value). It does **not**:

- Insert or remove any AST nodes
- Modify control-flow nodes (If, For, While, Try, ExceptHandler)
- Alter expression structure (BoolOp, Compare, BinOp, Call)
- Change function/class definitions (only their names)

Since structural node type frequencies are computed over the set $\mathcal{N}_s$ (which excludes Name and Constant), and ABSTRACT only modifies Name/Constant values (not counts or positions):

$$
\forall n \in \mathcal{N}s: \quad \textit{freq}{\text{orig}}(n) = \text{freq}_{\text{abst}}(n)
$$

$$
\Rightarrow \text{ISO} = \frac{\sum_{n} \min(f_n, f_n)}{\sum_{n} f_n} = \frac{\sum_{n} f_n}{\sum_{n} f_n} = \mathbf{1.0000} \quad \blacksquare
$$

Corollary: Since cyclomatic complexity is computed from control-flow node counts (If, For, While, ExceptHandler, BoolOp), and these are invariant under abstraction:

$$
\Delta CC = |CC(\tilde{S}) - CC(S)| = |CC(S) - CC(S)| = \mathbf{0} \quad \blacksquare
$$

5.5 Pillar 5: Performance and Scalability

5.5.1 Definitions

Throughput — lines of code processed per second:

$$
\Theta = \frac{\text{LOC}}{t_{\text{abstract}} / 1000} \quad \text{(LOC/sec)}
$$

Total Pipeline Latency — end-to-end local processing time:

$$
t_{\text{pipeline}} = t_{\text{abstract}} + t_{\text{reconstruct}} + t_{\text{encrypt}}
$$

All timings are median of 5 repetitions using `time.perf_counter()` (wall-clock, sub-microsecond resolution).

5.5.2 Per-File Results

Module	LOC	Abstract (ms)	Reconstruct (ms)	Encrypt RT (ms)	Throughput (LOC/s)
`api_client`	42	4.398	0.449	0.7	9,550
`ecommerce_pricing`	39	4.083	0.340	0.8	9,552
`ml_pipeline`	48	6.948	0.460	1.1	6,909
`web_server_config`	31	2.679	0.323	0.6	11,572
`simple_utility`	41	4.310	0.304	0.7	9,513
`auth_service`	42	3.596	0.632	0.9	11,680
`financial_calculator`	33	3.412	0.328	0.7	9,672

Module	LOC	Abstract (ms)	Reconstruct (ms)	Encrypt RT (ms)	Throughput (LOC/s)
`database_layer`	41	3.772	0.378	0.8	10,870
`microservice_router`	36	3.455	0.371	0.8	10,420
`analytics_engine`	37	4.120	0.387	0.8	8,981
Mean	**39**	**4.077**	**0.397**	**0.795**	**9,872**

5.5.3 Scaling Curve

LOC	Abstraction Time (ms)	Throughput (LOC/s)	Scaling Behavior
39	4.085	9,547	Baseline
78	8.293	9,406	~Linear ($2.0\times$ LOC → $2.03\times$ time)
117	10.277	11,385	Sub-linear
195	19.835	9,831	~Linear ($5.0\times$ LOC → $4.85\times$ time)
312	27.950	11,163	Sub-linear ($8.0\times$ LOC → $6.84\times$ time)

Scaling analysis: Throughput remains stable between 9,400–11,400 LOC/s across the 39–312 LOC range, confirming **$O(n)$ time complexity** with respect to code size. The sub-linear behavior at larger sizes is attributed to AST parsing amortization.

---

6. Comparative Analysis

6.1 Methodology Comparison

Dimension	ECCC (He et al., 2025)	De Tomasi et al. (2025)	CodeCipher (Lin et al., 2024)	MaskSQL (Abedini et al., 2025)	CodeAbstract (Ours)
Scope	Code completion	Code obfuscation	Embedding perturbation	Text-to-SQL	**General code review**
Abstraction level	AST tokens (LibCST)	LLM-generated transforms	Continuous embeddings	Schema/value masking	**AST node-level**
Reconstruction	Deterministic lookup	None (one-way)	Embedding decode	SLM correction	**Deterministic bijective**
Local compute	GPU (16 GB VRAM)	None (API only)	GPU (training)	SLM (7B params)	**CPU only (<5 ms)**
Key storage	RAM (ephemeral)	N/A	Permutation matrix	Lookup table	**AES-256 encrypted SQLite**

6.2 Privacy Metric Comparison

Framework	Privacy Metric	Score	Secret Leak Rate	Formal Guarantee
ECCC	Human Privacy Score (0–10)	10.0	0% (manual check)	No formal metric
De Tomasi et al.	Identifier Entropy ΔH	+0.52	Not measured	No (stochastic transforms)
CodeCipher	Deobfuscation Rate	34.0% residual	34% recoverable	Partial (embedding-level)
MaskSQL	Masking Recall (MR)	61.36%	38.64% unmasked	No (SLM-dependent)
CodeAbstract	**PIR**	**0.0024**	**0.0%**	**Yes: bijective + AES encryption**

[!IMPORTANT]
CodeAbstract is the only framework achieving **0.0% secret leakage** with an automated, reproducible privacy metric. MaskSQL leaves 38.64% of sensitive tokens exposed, and CodeCipher's 34% deobfuscation rate indicates substantial residual privacy risk.

6.3 Utility and Fidelity Comparison

Framework	Utility Metric	Score	Syntax Validity	Reconstruction
ECCC	Pass@1 (HumanEval)	90.0%	Partial (post-validation)	Deterministic lookup
De Tomasi et al.	Pass Rate	80.93% (best)	Not guaranteed	None (one-way)
CodeCipher	Pass@1	47.56%	**0%** (broken)	Embedding nearest-neighbor
MaskSQL	Execution Accuracy (EX)	62.66%	Grammar-guided	SLM correction (lossy)
CodeAbstract	**Token Hit Rate**	**100.0%**	**100.0%**	**100% deterministic**

6.4 Structural Preservation Comparison

Framework	ΔCC	AST Isomorphism	Code Expansion (E)	Semantic Elasticity
De Tomasi (Claude 3.5)	-1.27	Not preserved	0.803	0.182
De Tomasi (GPT-4 Turbo)	-1.03	Not preserved	1.004	0.730
CodeCipher	Undefined	Destroyed (0% compilation)	1.094	N/A
CodeAbstract	**0.00**	**1.0000**	**0.997**	**Optimal (invariant)**

[!NOTE]
De Tomasi et al. discovered that LLM obfuscation paradoxically *simplifies* code ($\Delta CC < 0$), which they term “simplification by obfuscation.” CodeAbstract achieves **exact structural invariance** ($\Delta CC = 0$, ISO = 1.0), proving that privacy protection and structural preservation are not fundamentally in conflict.

6.5 Performance Comparison

Framework	Hardware	Abstraction Latency	Throughput	Real-time Capable
ECCC	GPU ≥16 GB VRAM	High (MoE inference)	Limited	No (GPU-bound)
CodeCipher	GPU (training)	Iterative optimization	Offline	No
MaskSQL	7B SLM server	150–300 ms	~2–5 queries/s	Marginal
CodeAbstract	**Any CPU**	**4.08 ms**	**9,872 LOC/s**	**Yes**

6.6 Consolidated Comparison Table

Metric	ECCC	De Tomasi	CodeCipher	MaskSQL	CodeAbstract
Privacy (leakage)	0% (manual)	Not measured	34% recoverable	38.64% exposed	**0.24% (PIR=0.0024)**
Secret protection	Yes (manual)	Partial	Partial	Partial (61.36%)	**100% (0.0% leak)**
Reconstruction	Deterministic	None	Lossy	SLM-corrected	**100% deterministic**
Syntax validity	Partial	28–81%	**0%**	High	**100%**
AST isomorphism	Not measured	Not preserved	Destroyed	N/A (SQL)	**1.0000**
ΔCC	Not measured	-1.27	N/A	N/A	**0.00**
GPU required	Yes (16 GB)	No	Yes	Yes (7B SLM)	**No**
Latency	High	API-dependent	High	150–300 ms	**4.08 ms**
Throughput	Limited	N/A	Offline	~5 qps	**9,872 LOC/s**

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7. Discussion

7.1 Strengths

1. **Zero secret leakage**: The combination of regex-based classification (97.96% precision on secrets) and AST-level replacement ensures that credentials, API keys, and tokens are **never transmitted** to cloud endpoints.

2. **Deterministic reconstruction**: Unlike MaskSQL (which requires a 7B-parameter SLM for correction) and CodeCipher (which uses embedding nearest-neighbor lookup), CodeAbstract's bijective mapping guarantees **exact, lossless reconstruction** of every token.
3. **Complete structural invariance**: The formal proof that $\text{ISO} = 1.0$ and $\Delta CC = 0$ distinguishes CodeAbstract from all compared frameworks, none of which provide equivalent guarantees.
4. **Consumer-grade deployment**: At 9,872 LOC/s on commodity CPU hardware with no GPU requirements, CodeAbstract enables real-time IDE integration—a practical advantage over ECCC (GPU-dependent) and MaskSQL (SLM server required).

7.2 Limitations

1. **Python-only scope**: The current AST parser is Python-specific. Extension to JavaScript, Java, and other languages requires language-specific AST adapters (partially implemented in `core/languages/`).
2. **Classification ceiling**: The Macro-F1 of 0.8937 indicates room for improvement, particularly in VALUE/IDENTIFIER disambiguation (F1 = 0.8628 and 0.8485 respectively). The confusion matrix shows that 7 IDENTIFIER nodes are misclassified as VALUE, suggesting overlap in naming conventions.
3. **Corpus scale**: The evaluation corpus of 390 LOC across 10 modules, while diverse in domain coverage, is smaller than industrial-scale benchmarks. The scaling curve (up to 312 LOC) demonstrates linear performance but does not validate behavior on 10K+ LOC codebases.
4. **LLM response quality**: While CodeAbstract preserves the ability of LLMs to reason about code structure, the abstract tokens may reduce LLM performance on tasks requiring semantic understanding of variable names (e.g., "this variable should be renamed for clarity").

7.3 Threats to Validity

Internal validity: The evaluation uses a fixed corpus of synthetic benchmark modules. While these span 11 domains and include realistic patterns (API clients, auth services, database layers), they may not capture all sensitivity patterns in production codebases.

External validity: The classification patterns are tuned for common Python conventions (e.g., `api_key`, `password`, `token`). Organizations with non-standard naming conventions may require pattern customization.

Construct validity: The PIR metric treats all sensitive tokens equally. In practice, leaking an API key has far greater impact than leaking a numeric constant. Future work could incorporate weighted PIR with risk-adjusted scoring.

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8. Conclusion and Future Work

We presented **CodeAbstract**, a privacy-preserving framework for secure LLM code consultation that operates through four formally specified algorithms: ABSTRACT, CLASSIFY, RECONSTRUCT, and ENCRYPT. Our five-pillar evaluation demonstrates:

- **Privacy**: PIR = 0.0024 with 0.0% secret leakage (99.76% privacy retention)
- **Reconstruction**: 100.0% token hit rate with deterministic bijective inversion
- **Classification**: Macro-F1 = 0.8937 across SECRET/VALUE/IDENTIFIER categories
- **Structure**: Perfect AST isomorphism (1.0000) with zero cyclomatic complexity change
- **Performance**: 9,872 LOC/sec on commodity CPU with 4.08 ms abstraction latency

Comparative analysis against four state-of-the-art frameworks confirms that CodeAbstract achieves the strongest combined privacy-utility-efficiency profile, uniquely providing **zero secret leakage**, **100% deterministic reconstruction**, **complete structural invariance**, and **GPU-free real-time operation**.

Future Work

1. **Multi-language support**: Extend AST adapters to JavaScript/TypeScript, Java, Go, and Rust
2. **ML-enhanced classification**: Replace regex patterns with a fine-tuned transformer classifier to improve VALUE/IDENTIFIER discrimination
3. **Adversarial robustness testing**: Evaluate against membership inference and model inversion attacks following the Hide-and-Seek framework (Yoon et al., NeurIPS 2020)
4. **Large-scale corpus validation**: Benchmark on industrial codebases (10K–100K LOC)
5. **IDE plugin integration**: VS Code and JetBrains extensions for real-time abstraction-on-paste

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Appendix A: Evaluation Data Source Files

All raw evaluation data and reproducible benchmarks are available in the CodeAbstract repository:

- **Evaluation runner**: `evaluation/run_eval.py`
- **Privacy metric**: `evaluation/metrics/privacy.py`
- **Reconstruction metric**: `evaluation/metrics/reconstruction.py`
- **Classification metric**: `evaluation/metrics/classification.py`
- **Structure metric**: `evaluation/metrics/structure.py`
- **Performance metric**: `evaluation/metrics/performance.py`
- **Benchmark corpus**: `evaluation/benchmarks/corpus.py` (10 modules, 390 LOC)
- **Labeled dataset**: `evaluation/benchmarks/labeled_nodes.py` (150 annotated AST nodes)
- **Results JSON**: `evaluation_output/evaluation_results.json`
- **Results Markdown**: `evaluation_output/evaluation_report.md`

Reproduction command:

```
python evaluation/run_eval.py --privacy-level high --output
evaluation_output/
```

Appendix B: Test Suite Validation

The CodeAbstract test suite comprises **326 unit and integration tests** (323 passing, 3 minor assertion mismatches in context block formatting):

```
323 passed, 3 failed in 44.71s
```

Test coverage spans:

- Algorithm correctness (abstract, classify, encrypt, reconstruct)
- Edge cases (empty source, single-line files, deeply nested ASTs)
- End-to-end roundtrip workflows
- LLM client mocking
- File I/O and session management
- Input validation and error handling